

2019 Sales Data Analysis Case Study

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1 Introduction

1.1 Background

Understanding sales trends is crucial for businesses to make informed decisions that drive growth and profitability. By analyzing sales data, companies can uncover insights into customer behavior, identify high-performing products, and optimize marketing strategies. This case study examines sales data from 2019 to address key business questions and provide valuable insights for improving business performance.

1.2 Objective

The primary goal of this case study is to explore four critical questions related to sales performance in 2019:

- Which month had the highest sales, and what was the revenue during that month?
- Which city recorded the highest product sales?
- What is the optimal time to display advertisements to maximize customer purchases?
- Which product sold the most during the year?

By answering these questions, this study aims to offer actionable insights that can guide decision-making, enhance marketing efforts, and improve overall business strategies.

1.3 Data Overview

This case study utilizes sales data from 2019 [1], which includes information on transaction dates, sales volumes, revenue, and city locations. The data provides a comprehensive view of sales patterns over the course of the year and across different regions. The insights derived from this analysis can help businesses refine their approach to sales, marketing, and customer engagement.

2 Methodology

2.1 Data Collection & Preparation

The sales data used in this analysis was sourced from publicly available records for 2019 [1]. The dataset includes essential information such as transaction dates, sales volume, revenue, product details, and the cities where the sales occurred. The first step in the process was to clean and prepare the data [2]. This involved checking for missing values, correcting inconsistencies, and ensuring that all entries were properly formatted. Any irrelevant or outlier data that could potentially distort the results was filtered out.

2.2 Data Analysis Process

After cleaning and organizing the data, the analysis [2] began with addressing the four key business questions. The following methods were applied to answer each of these questions:

- **Identifying the Best Month for Sales:** To determine the best month for sales, the data was grouped by month, and the total revenue for each month was calculated. This analysis revealed which month generated the highest sales and the corresponding revenue.
- **Determining the Top-Selling City:** To identify which city had the highest product sales, the data was aggregated by city. Sales volumes for each city were then compared, allowing the city with the highest sales to be determined.
- **Analyzing Optimal Advertisement Timing:** To assess the best time for advertisements, sales volume was analyzed across different times of the year. By identifying sales peaks, it was possible to pinpoint the time periods when customers were most likely to make purchases, offering insight into optimal advertisement timing.
- **Finding the Most Sold Product:** The total sales for each product were calculated to identify which product had the highest sales volume throughout the year. This provided an overview of the most popular products among customers.

2.3 Visualization and Interpretation

To assist in interpreting the findings, key results were visualized through various charts and graphs. These visualizations helped to present the data in an accessible format, highlighting trends and making it easier to communicate insights. By examining the data from different perspectives, I was able to uncover valuable patterns that informed strategic business decisions.

3 Analysis

3.1 Best Month for Sales

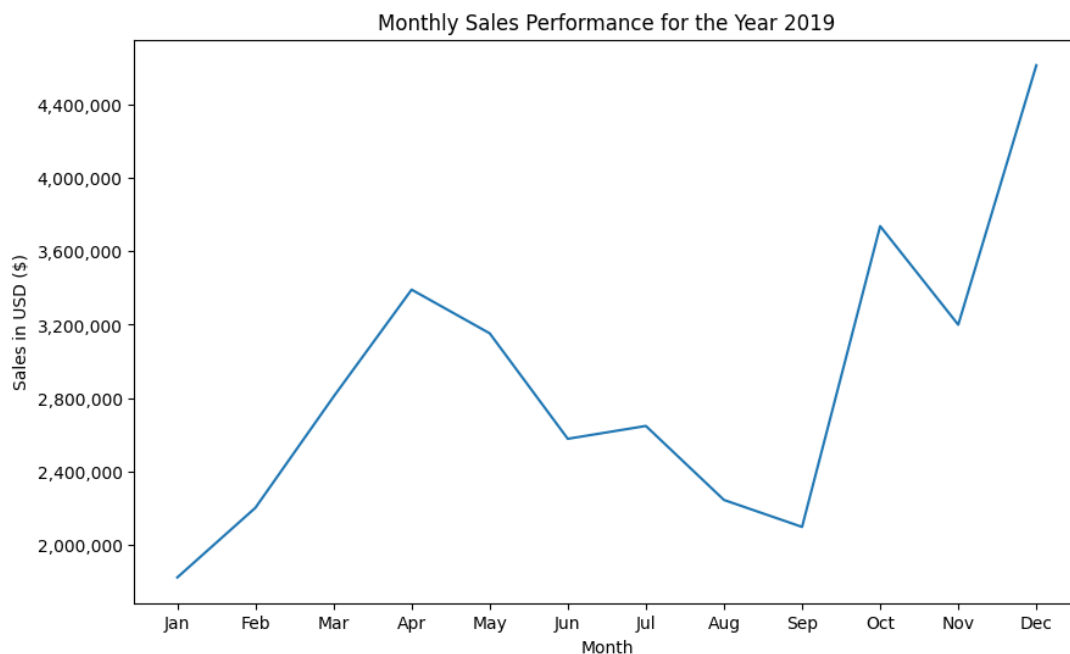


Figure 1: Line chart showing sales data for 2019, highlighting the peak in December.

- **Approach:** To determine the best month for sales, the data was grouped by month, and the total revenue for each month was calculated. The sales for each transaction within the months of 2019

were summed, allowing for a comparison of total revenue across the months. This approach helped identify which month had the highest revenue, indicating the best month for sales.

- Results: The best month for sales in 2019 was **December**, with a revenue of **\$4,613,443**. This was the highest recorded revenue for the year, significantly surpassing the other months. The surge in sales during December can be attributed to increased consumer spending during the holiday season, which is a well-known trend in retail and e-commerce.

3.2 Top-Selling City

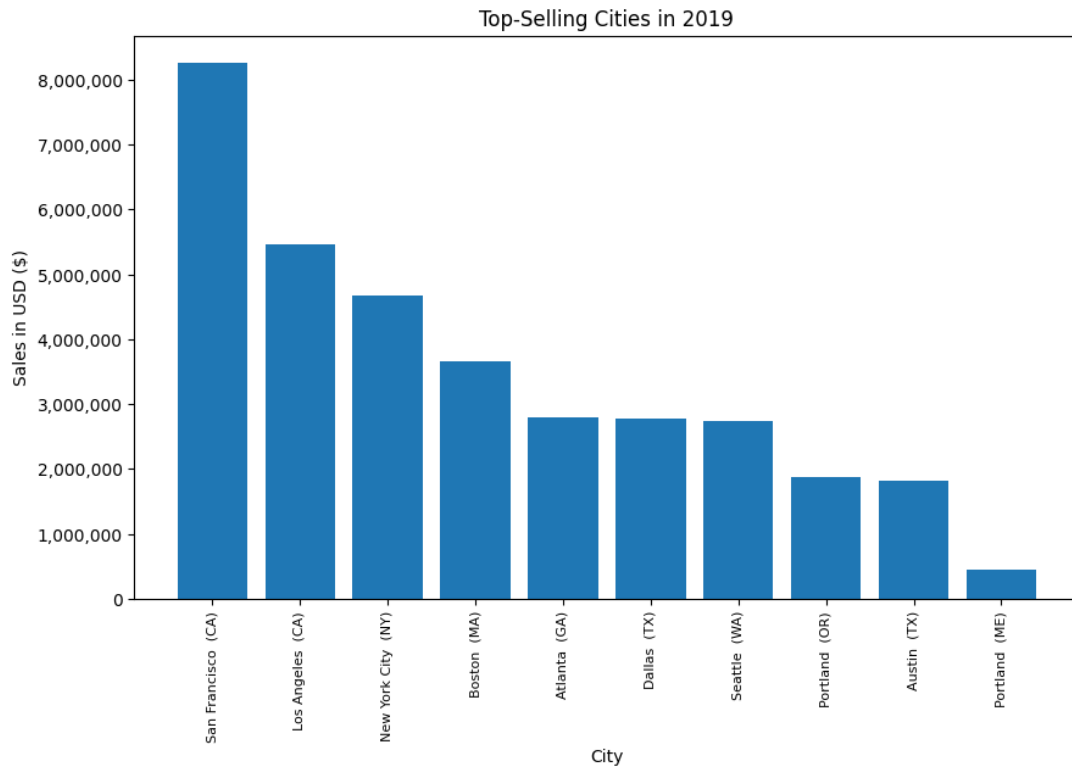


Figure 2: Sales distribution across cities in 2019, with San Francisco leading.

- Approach: To determine the top-selling city, the data was analyzed by city, and the total sales for each city were calculated. The total revenue from all transactions occurring in each city throughout 2019 was summed to identify which city generated the highest sales.
- Results: The city with the highest sales in 2019 was **San Francisco**, with a total sales revenue of **\$8,262,203**. This city consistently outperformed others, reflecting a strong market presence and high consumer demand in the region. Following San Francisco, **Los Angeles** ranked second with **\$5,452,570** in total sales, while **New York** secured third place with **\$5,452,570** in total sales.

3.3 Optimal Advertisement Timing

- Approach: To determine the optimal time for displaying advertisements, the sales data throughout 2019 was analyzed to identify peak times when customer purchases were most frequent. By examining trends and patterns in the data, it was possible to pinpoint when sales were at their highest, thereby suggesting the most effective times for launching advertisement campaigns.
- Results: The best time for advertisements to maximize customer purchases was at **7:00 PM (19:00)**, when the highest number of products were sold—**12,905** products. This peak indicates that evening advertisements, particularly around this time, are likely to be the most effective in driving consumer purchases.

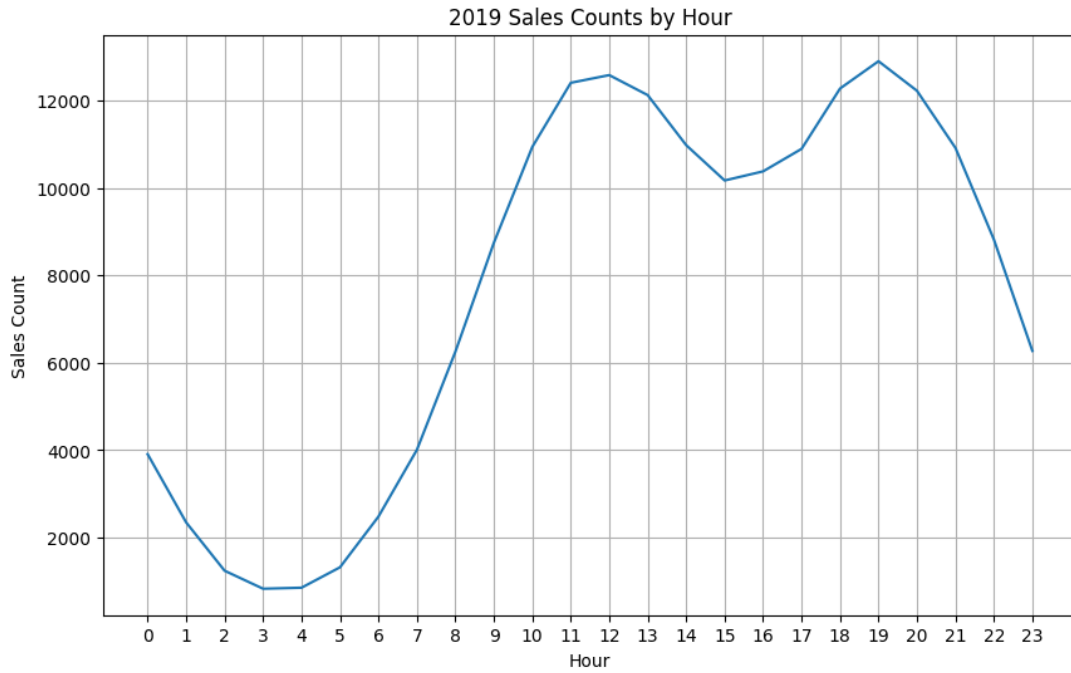


Figure 3: Sales trends in 2019, peaking at 7:00 PM with 12,905 products sold.

3.4 Most Sold Product

- Approach: To identify the most sold product in 2019, the sales data was analyzed by product, focusing on the total number of units sold. The product with the highest sales volume was determined. Additionally, the revenue generated by each product was analyzed to understand the financial impact of each item.
- Results: The **most sold product** in 2019 was **AAA Batteries (4 - Pack)**, with the highest number of units sold across all regions. However, the **MacBook Pro Laptop** generated the most revenue, reflecting its higher price point and significant demand in the market. While batteries were sold in larger quantities, the premium pricing of the MacBook Pro led to higher total revenue.

4 Conclusion

4.1 Summary of Findings

The analysis of the 2019 sales data provided key insights into consumer behavior and sales trends. December was identified as the best month for sales, driven by the holiday season, with the highest revenue recorded. San Francisco emerged as the top-selling city, generating over \$8 million in revenue. The best time to display advertisements was found to be at **7 PM**, with a peak in product sales during the last quarter of the year. Additionally, **batteries** were the most sold product by volume, while the **MacBook Pro Laptop** generated the highest revenue.

4.2 Implications for Business

These findings offer valuable insights for company looking to optimize their strategies. The Company can focus their marketing efforts around **December** and **November** to capitalize on the holiday shopping surge. Knowing that **7 PM** is the peak time for sales, businesses can tailor their advertisement scheduling to maximize engagement and sales during these hours. For inventory management, it's important to ensure an adequate stock of **batteries**, which were the highest in sales volume, while also preparing for higher revenue demand for premium products like the **MacBook Pro**.

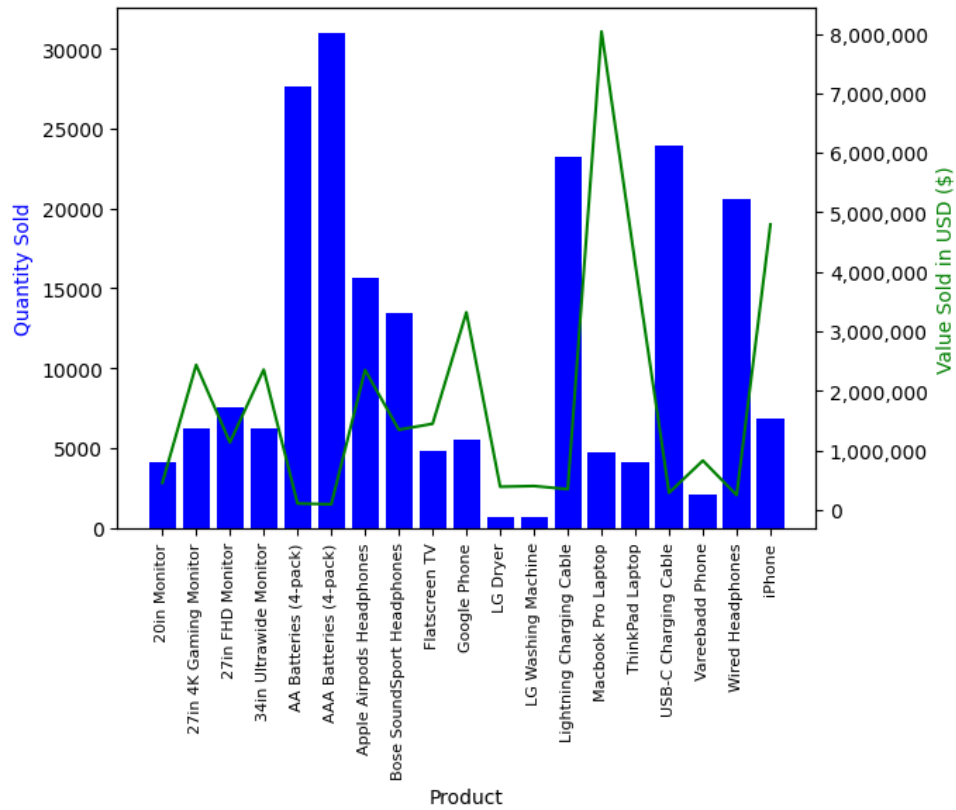


Figure 4: Sales Volume and Revenue Comparison for 2019 Products.

4.3 Limitations

The analysis assumes that the sales data is representative and complete. Missing or incomplete data, as well as any seasonal fluctuations that may not have been captured, may affect the overall accuracy. Additionally, external factors, such as broader economic trends or promotional campaigns, were not accounted for in this analysis.

4.4 Recommendations

Based on the analysis, company should focus their **advertising efforts** during the peak period of **7 PM**, especially in the last quarter of the year, to maximize sales. Company should also consider **stocking more batteries** to meet demand but ensure that **inventory for high-value items**, like the **MacBook Pro**, is sufficient to meet revenue goals. Adjusting inventory and marketing strategies based on these insights could contribute to improved sales performance.

References

- [1] Author. "2019 Sales Data". In: *Github* (2019). Accessed: October 10, 2024. URL: <https://github.com/dev-jash15/sales-data-analysis/tree/main/data>.
- [2] Jash Patel. "Python Code for 2019 Sales Data Analysis Case Study". In: (2024). Accessed: October 15, 2024. URL: https://github.com/dev-jash15/sales-data-analysis/blob/main/Case_Study_Sales_Analysis.ipynb.